

Bogyeom Park

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EDUCATION

Seoul National University of Science and Technology

Ph.D. in Applied Artificial Intelligence

- Advisor: Kyoungwon Seo

Seoul, South Korea

Mar. 2023 - Present

Seoul National University of Science and Technology

B.S. in Electrical and Information Engineering

Seoul, South Korea

Mar. 2019 - Feb. 2023

HONORS & AWARDS

- AI SeoulTech Graduate Scholarship (KRW 10,000,000), Seoul Scholarship Foundation, 2025
- Best Paper Award (co-author), HCI Korea 2025
- Best Student Paper Bronze Award, IEEE Seoul Section, 2024
- National Science and Engineering Undergraduate Scholarship (full tuition), Korea Student Aid Foundation, 2021-2022
- Best Capstone Design Award, Seoul National University of Science and Technology, 2022

JOURNAL ARTICLES

Integrating Biomarkers From Virtual Reality and Magnetic Resonance Imaging for the Early Detection of Mild Cognitive Impairment Using a Multimodal Learning Approach: Validation Study

Bogyeom Park, Yuwon Kim, Jinseok Park, Hojin Choi, Seong-Eun Kim, Hokyoung Ryu, and Kyoungwon Seo

Journal of Medical Internet Research, 26, e54538 (2024) - JCR Q1; JIF 8.2; 96th percentile, rank 8/194 in Health Care Sciences & Services (2026 JCR)

INTERNATIONAL CONFERENCE PAPERS

Assessing Critical Thinking through a Multi-Agent LLM-Based Debate Chatbot

Bogyeom Park and Kyoungwon Seo

CHI EA '25: Extended Abstracts of the 2025 CHI Conference on Human Factors in Computing Systems - Late-Breaking Work (32.7% acceptance; 619/1,888)

How Self-Disclosing Chatbots Influence Student Engagement, Assessment Accuracy, and Self-Reflection in Academic Stress Assessment

Minyoung Park, **Bogyeom Park**, and Kyoungwon Seo

CHI EA '25: Extended Abstracts of the 2025 CHI Conference on Human Factors in Computing Systems - Late-Breaking Work (32.7% acceptance; 619/1,888)

A Self-Determination Theory-Based Career Counseling Chatbot: Motivational Interactions to Address Career Decision-Making Difficulties and Enhance Engagement

Hyerim Han, **Bogyeom Park**, and Kyoungwon Seo

CHI EA '25: Extended Abstracts of the 2025 CHI Conference on Human Factors in Computing Systems - Late-Breaking Work (32.7% acceptance; 619/1,888)

Exploring the Multimodal Integration of VR and MRI Biomarkers for Enhanced Early Detection of Mild Cognitive Impairment

Bogyeom Park, Yuwon Kim, Jinseok Park, Hojin Choi, Seong-Eun Kim, Hokyoung Ryu, and Kyoungwon Seo

CHI EA '24: Extended Abstracts of the 2024 CHI Conference on Human Factors in Computing Systems - Late-Breaking Work (33.9% acceptance; 391/1,154)

Multimodal Machine Learning Model for MCI Detection Using EEG, MRI and VR Data

Mariem Kallel, **Bogyeom Park**, Kyoungwon Seo, and Seong-Eun Kim

2024 International Technical Conference on Circuits/Systems, Computers, and Communications (ITC-CSCC), pp. 1-6

Advancing Mild Cognitive Impairment Detection: Integrating VR, MRI, and Neuropsychological Insights for Comprehensive Diagnosis

Bogyeom Park, Jinseok Park, Hojin Choi, Hokyoung Ryu, and Kyoungwon Seo

2024 International Technical Conference on Circuits/Systems, Computers, and Communications (ITC-CSCC), pp. 1-6

Early Screening of Mild Cognitive Impairment Using Multimodal VR-EP-EEG-MRI (VEEM) Biomarkers via Machine Learning

Se Young Kim, **Bogyeom Park**, Dohyun Kim, Hojin Choi, Jinseok Park, Hokyoung Ryu, and Kyoungwon Seo

2024 International Conference on Electronics, Information, and Communication (ICEIC), pp. 1-4

KOREAN CONFERENCE PAPERS

From Teacher Needs to Agentic AI: Designing and Validating a Personalized Career Counseling System

Bogyeom Park, Mina Yoo, Dongkuk Lee, Mi-ae Choi, Seona Park, So Young Jo, and Kyoungwon Seo

Proceedings of HCI Korea 2026 - Oral Presentation

Counterfactual vs. Prefactual: Two Narrative AIs Improve Causability for Health Data by Different Mechanisms

Hyobin Park (now **Bogyeom Park**) and Kyoungwon Seo

Proceedings of HCI Korea 2023, pp. 828-835

RESEARCH EXPERIENCE

Human-centered Artificial Intelligence (HAI) Lab, SeoulTech

Seoul, South Korea

Graduate Researcher (Advisor: Kyoungwon Seo)

Mar. 2023 - Present

Undergraduate Researcher

Mar. 2021 - Feb. 2023

ICAP-Based AI Tutoring System for Probability and Statistics Learning

Lead Graduate Researcher | Tutorus Labs

Mar. 2026 - Aug. 2026

- Led the design and evaluation of an ICAP-based AI tutor that uses staged elicitation to promote active, constructive, and interactive engagement rather than answer delivery
- Developed an utterance-level coding and analytics workflow linking dialogue evidence with tutor correctness, usage patterns, and learning outcomes
- Built an interactive research dashboard for reviewing engagement labels and comparing AI-tutored learning processes

Agentic AI for Personalized Career, Academic, and Counseling Support

Lead Graduate Researcher | Korea Education & Research Information

Jun. 2025 - Dec. 2025

Service

- Led a study of how agentic AI could support personalized educational pathways, counseling, and decision-making

AI Copilot Technologies for Adaptive, Teacher-Augmented Learning

Lead Graduate Researcher | Institute for Information & Communications

Jul. 2023 - Dec. 2025

Technology Planning & Evaluation

- Designed and evaluated a multi-agent LLM debate chatbot that draws out a student's argument and assesses critical thinking from the exchange
- Co-developed a self-determination-theory career counseling chatbot that addresses career decision-making difficulties through motivational dialogue

AI-Based Early Screening and Prognosis Prediction for Landscape Tree Disease

Lead Graduate Researcher | SUNGHA Co., Ltd.

Jul. 2023 - Dec. 2023

- Built an expert-validated image dataset of Zelkova serrata, the species that accounts for more than half of Korea's legally protected trees
- Compared transfer-learning models with and without plant-disease pre-training for early screening of tree disease

VR-EP-EEG-MRI Digital Biomarker Basic Research Laboratory

Lead Graduate Researcher | National Research Foundation of Korea

Mar. 2022 - Feb. 2024

- Ran the VR data collection on site at Hanyang University Guri Hospital with patients with mild cognitive impairment and healthy controls
- Analyzed the resulting VEEM multimodal dataset - VR interaction, eye tracking, evoked potentials, EEG and MRI - including the VR-MRI-SNSB cohort of 54 participants

Multimodal Deep Learning for Early Dementia Diagnosis from VR Daily-Living Data

Lead Graduate Researcher | National Research Foundation of Korea

Mar. 2021 - Feb. 2024

- Developed the virtual kiosk task in Unity - the six-step daily-living scenario the study's VR biomarkers are derived from
- Led the development and validation of a multimodal model combining VR behavioral biomarkers with MRI for early detection of mild cognitive impairment
- Derived hand-movement and gaze features from the kiosk task and tested them against MRI-derived measures in a clinical sample

TEACHING & MENTORING

2026 AX Academy Big Data Boot Camp

May 2026 - Aug 2026

Tutor, Hyundai Motor Group

- Mentored nine participants from across the company, each developing an individual big data project from proposal to final deliverable
- Advised on analysis design and revised project plans with participants through the boot camp's project-based sprints

Deep Learning

Fall 2023

Teaching Assistant, Seoul National University of Science and Technology

- Designed a final project using CNN-based models to predict drivers' physical and cognitive states from image data
- Supported lectures, advised student projects, and graded assignments and examinations

SKILLS

- **Research Methods** - experimental design, usability evaluation, user research, prototyping, multimodal learning analytics, statistical analysis, qualitative coding
- **Machine Learning** - PyTorch, scikit-learn, Hugging Face, feature engineering, multimodal fusion
- **LLM Systems** - OpenAI and Anthropic APIs, multi-agent pipelines, RAG, prompt engineering and evaluation, LLM-as-a-judge
- **Programming and Tooling** - Python, C/C#, Docker, Streamlit, Git, SPSS
- **VR and Interactive Systems** - Unity, VR task development, interaction logging and eye tracking
- **Design** - HCI theory, UX design, interaction prototyping, usability testing

PATENT

Explainable AI-Based System for Early Diagnosis and Prognosis Prediction of Alzheimer's Disease Using VR Biomarkers, and Method Thereof (Korean Patent Application No. 10-2023-0105821, filed Aug. 2023)

ACADEMIC SERVICE

- Student Volunteer, AIED 2026
- Publicity Manager, SeoulTech Human-centered Artificial Intelligence Lab